Business Key-Scenarios

Proposed Use Case Framework

Primary Business Actors

User Type	Description
Radiologist	Reviews diagnostic images.
Cardiologist	Analyzes cardiac structures.
Imaging Technician	Produces imaging studies.
AI Platform	Executes AI workflows.
AI Governance Team	Monitors model quality.
Hospital IT	Operates the platform.

Macro Business Capabilities

Capability	Value
Imaging Acquisition	Collect imaging studies
AI-Assisted Analysis	Accelerate interpretation
3D Reconstruction	Improve visualization
Clinical Decision Support	Improve diagnostic confidence
Continuous Learning	Improve future performance

Scenario 1 — Secure Imaging Acquisition and Preparation

Business Goal

Reduce operational effort required to prepare imaging studies for advanced analysis while ensuring patient privacy and regulatory compliance.

Actors

- Imaging Technician
- PACS
- AI Platform

Current Challenges

Imaging studies are often:

- distributed across systems
- manually transferred
- inconsistently prepared
- subject to privacy concerns

These activities create delays and increase operational risks.

Future Vision

When a new imaging study becomes available, the platform automatically prepares it for downstream analysis while enforcing privacy controls and traceability requirements.

Business Value

- Reduced operational workload
- Faster processing initiation
- Improved compliance posture
- Reduced privacy exposure

Key Constraints

- GDPR compliance
- Hospital governance policies
- Data ownership requirements

AI Opportunity

Low. This scenario is primarily automation and governance driven.

Scenario 2 — AI-Assisted Anatomical Segmentation

Business Goal

Reduce the effort required to identify anatomical structures within complex imaging studies.

Actors

- Radiologist
- Cardiologist
- AI Platform

Current Challenges

Clinicians spend significant time manually reviewing large imaging datasets before performing analysis. This activity is repetitive and time consuming.

Future Vision

The platform automatically identifies relevant anatomical structures and presents preliminary segmentation results for clinical review.

Business Value

- Reduced analysis preparation time
- Increased consistency
- Improved clinician productivity

Key Constraints

- Human validation remains mandatory
- Results must be explainable
- Clinical accountability remains unchanged

AI Opportunity

Very High.
This represents one of the primary value-generating AI capabilities.

Scenario 3 — Automated 3D Reconstruction

Business Goal

Transform 2D imaging studies into navigable 3D anatomical representations.

Actors

- Cardiologist
- Surgeon
- AI Platform

Current Challenges

Understanding complex anatomical structures from 2D slices can be time-consuming and cognitively demanding.

Future Vision

Validated segmentation outputs are automatically transformed into 3D models suitable for advanced visualization environments.

Business Value

- Improved surgical planning
- Enhanced anatomical understanding
- Better collaboration among specialists

Key Constraints

- High computational requirements
- Accuracy requirements
- Clinical validation requirements

AI Opportunity

High.

Scenario 4 — Clinical Decision Support

Business Goal

Assist clinicians in identifying potentially relevant abnormalities.

Actors

- Radiologist
- Cardiologist
- AI Platform

Current Challenges

Large imaging volumes increase the possibility of oversight.

Future Vision

The platform highlights regions requiring additional attention and provides supporting evidence for clinician review.

Business Value

- Increased review consistency
- Reduced oversight risk
- Improved diagnostic confidence

Key Constraints

- No autonomous diagnosis
- Human oversight required
- Explainability required

AI Opportunity

Very High.

Scenario 5 — Human Validation and Governance

Business Goal

Ensure clinicians remain responsible for all diagnostic decisions.

Actors

- Radiologist
- Cardiologist
- Clinical Governance Team

Current Challenges

AI systems can create trust concerns if decision-making processes are not transparent.

Future Vision

Every AI-generated result is reviewed, accepted, modified, or rejected by qualified clinical personnel.

Business Value

- Clinical trust
- Regulatory compliance
- Adoption enablement

Key Constraints

- Full traceability
- Auditability
- Human accountability

AI Opportunity

Medium.
The value is primarily governance rather than intelligence.

Scenario 6 — Continuous Learning and Institutional Knowledge Growth

Business Goal

Transform clinician feedback into a strategic organizational asset.

Actors

- Clinicians
- AI Governance Team
- Data Science Team

Current Challenges

Expert knowledge generated during daily operations is rarely captured in reusable form.

Future Vision

Validated clinical feedback continuously improves organizational datasets and future AI capabilities.

Business Value

- Increasing AI performance
- Creation of proprietary knowledge assets
- Long-term competitive advantage

Key Constraints

- Governance approval
- Data quality controls
- Privacy requirements

AI Opportunity

Very High.

Business Capability Mapping

Capability	Scenarios
Privacy-Preserving Ingestion	S1
AI-Assisted Analysis	S2, S4
3D Visualization	S3
Clinical Governance	S5
Continuous Learning	S6